Diffusion Self-Distillation for Remote Sensing Scene Classification

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***Abstract*— Remote sensing scene classification, a fundamental task in remote image analysis, has obtained rapid progress due to the powerful capabilities of convolutional neural networks (CNNs). Achieving precise classification performance heavily relies on the feature extraction capacity of the network. However, due to the large variation and severe distortion within the images, extracting robust feature representations is necessary but challenging. Self-distillation could enhance the shallow layers by providing stronger gradients and more accurate supervision from deeper layers, thereby promoting the extraction of spatially detailed features. Nonetheless, due to the limited capacity of shallow layers to learn truly valuable knowledge, shallow layer features can be viewed as the noisy version of deep layer features and contain more disruptive factors, which significantly impedes the effectiveness of self-distillation. To address this issue, in this article, we establish the diffusion self-distillation network (DSDNet), which incorporates the conditional diffusion denoising model into the self-distillation framework. Specifically, DSDNet filters noise from shallow features through the diffusion denoising process, enabling more precise and accurate distillation between the refined student features and the teacher features. Extensive experiments on four challenging remote sensing datasets demonstrate that the proposed DSDNet achieves significant performance improvements over various backbone networks with negligible increases in parameters, delivering state-of-the-art classification performance. Our code is available on https://github.com/toggle1995/DSDNet**

***Index Terms*— Diffusion, knowledge transfer, remote sensing scene classification, self-distillation.**

Received 3 November 2024; revised 14 March 2025; accepted 14 April 2025. Date of publication 13 May 2025; date of current version 16 June 2025. This work was supported in part by the Start-Up Research Fund of Southeast University under Grant RF1028624156, in part by the Natural Science Foundation of Beijing Municipality under Grant 3252014, and in part by the Big Data Computing Center of Southeast University. *(Corresponding author: Xiaoyan Luo.)*

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Digital Object Identifier 10.1109/TGRS.2025.3569616

I. Introduction

RECENT advancements in various sensors and remote sensing technologies have greatly facilitated the acquisition of high-resolution images, which contain complex and rich spatial information. These developments enable more comprehensive analysis and spur the emergence of numerous subtasks toward precise remote sensing interpretation [1], [2], [3], [4], [5]. Among these, remote sensing scene classification has emerged as a fundamental and critical task, garnering significant attention due to its broad applications in urban management, emergency response, and security surveillance [6], [7]. However, considering the significant variations and disturbances within the images, remote sensing scene classification continues to pose formidable challenges [7], [8], [9].

As depicted in the first row of Fig 1, although these images belong to different classes, they exhibit similar visual characteristics. In contrast, as shown in the second row, although the four images come from the same class, i.e., "Church," their visual appearances are extremely different due to variations in shape, scale, structure, etc. As a result, to recognize different categories accurately, extracting discriminative and class-specific feature representations plays an important role in remote sensing scene classification [10].

Recently, with the rapid progress and great success of deep-learning-based methods, extracting features through the convolutional neural networks (CNNs) or transformer has become predominant in remote sensing scene classification. Various backbones, such as VGGNet [11], ResNet [12], and ViT [13], have been effectively used for this task. However, the conventional training pipeline typically applies supervision only at the final layer, leading to the gradient vanishing issue in the shallow layers, which impairs the extraction of spatial details from these layers [14], [15]. To mitigate this problem, strategies such as deep supervision and self-distillation have been proposed to enhance gradient propagation in the shallow layers [14], [16], [17], [18]. Deep supervision forces different layers to conform to the same learning target, which disrupts the connectivity and feature hierarchy established in CNNs [17], [19], [20]. Self-distillation avoids this issue through a progressive learning paradigm, where the deep layers of the network act as teachers to supervise their adjacent shallow layers. However, as depicted in Fig. 2, features from the shallow layers present the more disorganized distribution, which indicates greater perturbations compared with deep layer features. According to previous work [21], the effectiveness of distillation could be significantly impaired by the

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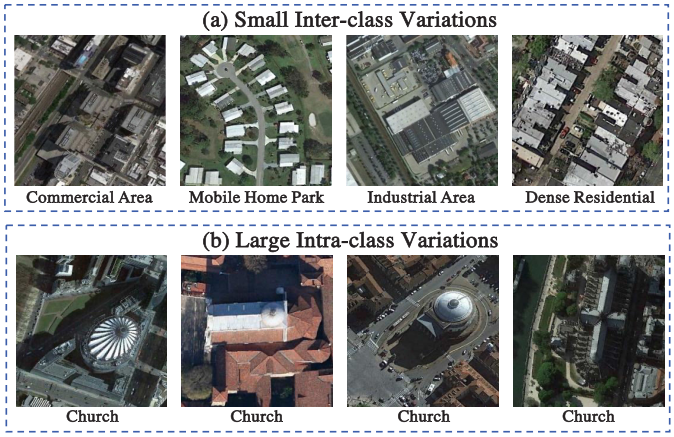


Fig. 1. (a) Images come from four different categories, yet they exhibit similar appearances. (b) Four images belong to the same category "Church." However, they display significantly different appearances.

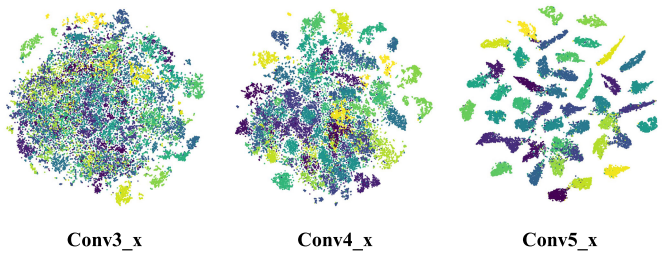


Fig. 2. t-SNE visualization of feature maps from the NWPU-RESISC45 dataset, extracted at three different stages of ResNet-50, shows that the shallower the layer, the more disorganized the extracted features.

noise within the student model. As a result, the disruptive factors in the shallow layer features also lead to undesired degradation in self-distillation. Particularly, this phenomenon is pronounced in the remote sensing scene classification due to the significant perturbations in remote sensing images caused by the low-light condition and weather interference.

In this article, to solve the problems mentioned above, we establish diffusion self-distillation network (DSDNet), which integrates the conditional diffusion model (DM) into the self-distillation. The overall architecture of our DSDNet is illustrated in Fig. 3. Basically, following the self-distillation network, DSDNet incorporates the classifier at different intermediate layers. Then, the classifier on the deeper layer works as the teacher to supervise its adjacent shallower counterpart. Namely, the shallow layer takes the prediction logits from the deeper layers as the learning targets. The logits could be regarded as the soft-label containing "dark knowledge" that is beneficial for the classification [22], [23]. In this way, DSDNet establishes the sequential supervision following the hierarchical structure of each layer, without disrupting the progressive learning paradigm of the neural network [17]. Furthermore, as advocated in [21], distilling knowledge in the presence of noise can be detrimental to the student model and may lead to degradation. This issue is more significant in self-distillation due to the limited capacity of shallow layers to learn valuable knowledge. To mitigate this problem, our DSDNet denoises the student features to generate refined representations for effective distillation. Prior study [7] observes that the features from the shallow layers are coarser than those from deeper layers. We further hypothesize that the features from the student (shallow) branch could be seen as noisy versions of those from the teacher (deep) branch. Then, inspired by the success of generative tasks [24], [25], we implement the conditional DMs to remove the noise stored in the student features. To this end, we design the diffusion distillation module (DiDi module) within DSDNet to filter the noise and distill the knowledge effectively. DiDi module connects two adjacent classifiers and takes their prediction logits as inputs, with the deeper one regarded as the teacher feature and the other as the student feature. In the diffusion-training process, Gaussian noise controlled by a variance schedule is added to the teacher feature. Then the DM $\Phi_\theta(\cdot)$ is trained to predict the noise intensity, conditioned on the student feature [26], [27]. Synchronously, alongside the progressive learning of $\Phi_\theta(\cdot)$, task-irrelevant noise in the shallow layer features could be removed via the step-by-step denoising during the diffusion-inference process. Subsequently, knowledge transfer is performed based on the refined student feature and the corresponding teacher feature. In this way, DSDNet enables more precise and effective distillation, thereby enhancing the capability of the shallow layers and ultimately improving classification accuracy.

Notably, it is essential to mention that the diffusion-training and diffusion-inference processes are conducted jointly during the training of DSDNet to refine the student feature and facilitate self-distillation. Meanwhile, the distillation, along with the diffusion process, is exclusively performed during the training of DSDNet. During the inference of DSDNet, all the auxiliary classifiers and the DiDi modules will be discarded, leaving only the final classifier to generate predictions. This ensures that compared with a plain network, DSDNet introduces no additional parameters or computational overhead during inference. Extensive experiments conducted across four challenging remote sensing scene classification datasets demonstrate that DSDNet operates in a plug-and-play manner, yielding significant performance improvements across various backbones. Moreover, we synthesize various distortions in remote sensing images, and DSDNet exhibits remarkable robustness with minimal performance degradation.

The contributions of this article are summarized as follows.

1) We propose DSDNet to address the disruptive factors present in the features from shallow layers, thereby enhancing the effectiveness of self-distillation. To the best of our knowledge, this is the first implementation that integrates a conditional DM into the self-distillation framework.
2) We design a diffusion distillation (DiDi) module to refine raw features from the shallow layers, improving the effectiveness of self-distillation. Specifically, a conditional DM is integrated within the DiDi module, synchronizing the diffusion-training and diffusion-inference processes. This synchronization allows the denoising process to occur simultaneously with the knowledge transfer inherent in self-distillation.
3) We conduct extensive experiments on four challenging remote sensing scene classification datasets. The experimental results demonstrate that our DSDNet significantly improves the performance of plain network with a negligible increase in parameters and establishes

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the new state-of-the art. Meanwhile, the experiments on distorted images demonstrate the great robustness of DSDNet.

The rest of this article is organized as follows. In Section II, we introduce the related work relevant to DSDNet. In Section III, the proposed DSDNet is described in detail. In Section IV, we present extensive experiments to verify the effectiveness of DSDNet. Finally, the conclusion is summarized in Section V.

II. Related Work

A. Remote Sensing Scene Classification

Over the past decades, numerous efforts have been made toward the remote sensing scene classification. To obtain satisfactory classification performance, extracting high-quality discriminative feature representations plays a critical role.

In the early years, hand-crafted feature descriptors, such as color histogram (CH) [28], scale-invariant feature transform (SIFT) [29], vector of locally aggregated descriptors (VLAD) [30], and bag-of-visual-words (BoVW) [31], are widely used in remote sensing scene classification [32], [33], [34], [35], [36]. However, designing these descriptors is time-consuming, and their discriminative capability is limited, which makes them less suitable to meet the demands of real-world applications.

To address the limitations of hand-crafted descriptors, CNNs have been widely used in remote sensing scene classification. Among these methods, attention mechanisms play a crucial role by localizing significant regions that contain discriminative clues. For example, Wan et al. [9] propose a lightweight plug-and-play module, which obtains attention weights based on mathematical calculation of feature maps and locates significant regions efficiently. CDLNet [37] decomposes global features into a set of critical semantic vectors to guide the attention learning process of the network, enabling the network to focus on salient regions of the image and thus extract more discriminate representations. Meanwhile, some methods generate discriminative representation via multiple feature fusion. MGSNet [38] uses contrast regularization method to separately extract target and background information, which are combined through self-guidance to enhance feature representation.

Moreover, the highly successful transformer-based networks have also been brought into remote sensing scene classification for feature extraction [8], [13], [39], [40]. EMTCAL [41] combines the advantages of CNN and transformer to mine information within the image, which uses the convolution operation to extract multilevel features and develop multiscale transformer block to acquire the relationship between various scales. HHTL [40] exploits both the local knowledge and global contextual relationship through a dual-branch structure that extracts homogeneous and heterogeneous information simultaneously. HFFT-PD [8] consists of a hierarchical transformer merging (HTM) block and a lightweight adaptive channel compression (LACC) module, bridging the semantic gaps between features from different stages and producing a discriminative feature representation.

In addition, to produce discriminative feature representations for remote sensing images, soft labels that encapsulate interclass similarity information are used to mitigate class entanglement problems [7]. Specifically, T-CNN [42] proposes an adaptive transfer learning framework for remote sensing scene classification, which leverages the model predictions to iteratively refine class probabilities and generates the soft label through a dynamic label smoothing strategy. In this way, the soft label could provide more information about interclass relationship and improves generalization capacity. CASD [43] deploys multiple branches to capture diverse classification information, and the class-aware knowledge is distilled into the student branch through a soft probability distribution. UACL [44] exploits the unreliable samples and designs a soft contrastive learning strategy to measure the similarity among samples, which improves the efficacy of contrastive learning via the interclass entanglement information. Besides, some works [7], [18] use the self-distillation strategy to learn soft-label adaptively and then the soft label is used to provide more precise supervision with rich interclass relationship. Specifically, ESD-MBENet [18] establishes an ensemble framework and learns the soft label according to the logits from each branch, based on which the interclass similarity could be exploited and then distilled into the main branch.

Furthermore, to promote the representational capacity of the shallow layers, some methods [7], [18], [45], [46] introduce auxiliary subbranches from intermediate layers to establish a hierarchical supervision framework, which facilitates the gradient flow and strengthens layerwise feature learning. For example, ESD-MBENet [18] constructs an ensemble network with multiple subbranches attached to intermediate layers, ensuring that the shallow parts of the network receive more sufficient supervision. VSDNet [7] applies supervision between adjacent intermediate layers, promoting hierarchical consistency and improving the capacity of the shallow layers. Similarly, FASDNet [46] establishes a feature-augmented self-distilled CNN, in which two auxiliary classifiers are constructed to provide additional supervision information for intermediate layers.

B. Knowledge Distillation

Knowledge distillation is first proposed by Hinton et al. for model compression [47], in which students are trained to mimic the prediction logits from the teacher model. During distillation, the logits from the teacher contain the "dark knowledge" that reflects the interclass similarity, which can effectively guide the feature learning of the student beyond mere compression purposes. Since its inception, various innovative distillation approaches have been developed to enhance knowledge transfer more effectively. For example, FitNet [48] involves the intermediate feature representation for knowledge transfer besides prediction logits. AT [49] leverages attention maps to facilitate effective distillation. In addition, SSKD [22] uses self-supervision tasks as auxiliary tools to enrich the knowledge transferred from the teacher network.

In the works discussed above, knowledge distillation is typically conducted between different networks, which increases

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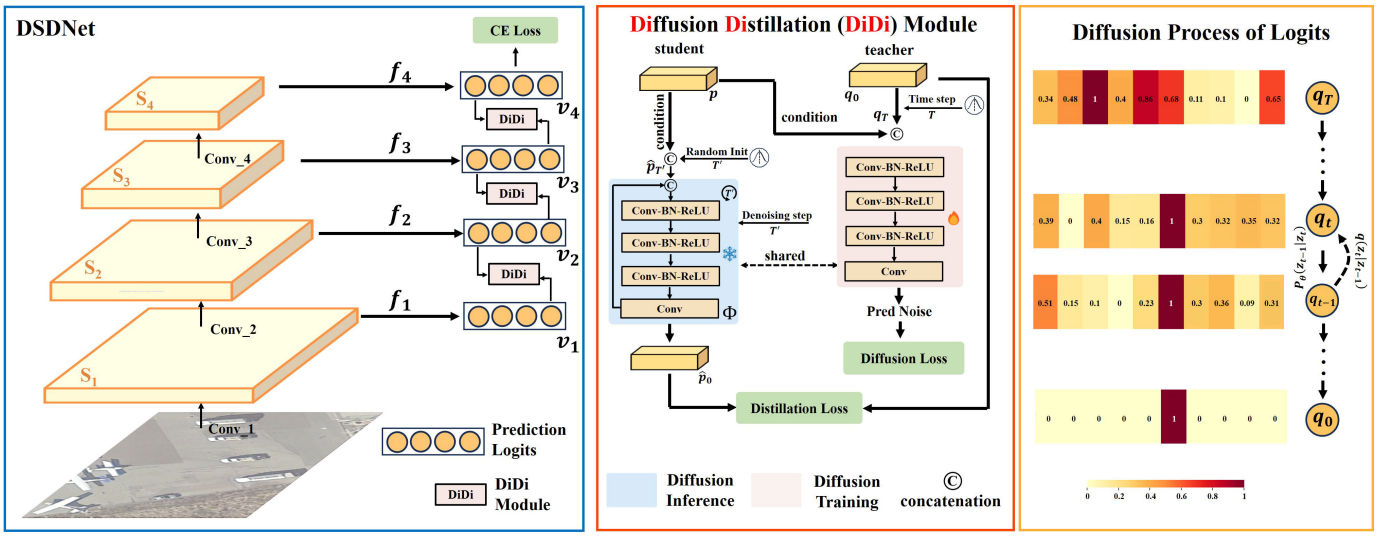


Fig. 3. Details of our DSDNet. The left part illustrates the structure of DSDNet. The backbone network is composed of several convolutional blocks $\{\text{Conv_1}, \text{Conv_2}, \ldots, \text{Conv_}M\}$ and generates feature maps $S = \{S_1, S_2, \ldots, S_M\}$. Obviously, $M = 4$ in the above figure. The middle part shows the details of DiDi module, in which the conditional DM Φ is used to remove the noise from the student logits. The right part illustrates the diffusion process of logits. Notably, the diffusion-training and diffusion-inference are jointly performed during the training of DSDNet to facilitate the distillation. During the inference of DSDNet, all the auxiliary classifiers (f_1, f_2, f_3) and DiDi modules will be discarded. Only the backbone network and f_4 will be maintained to produce the final prediction.

the training overhead due to the necessity of multiple models. To achieve the more compact and efficient distillation, self-distillation has been recently proposed, thereby reducing the reliance on multiple models and streamlining the training process. As the pioneer, Zhang et al. [16], [50] attach the classifiers to different intermediate points within a network, which are supervised by the prediction logits from the final classifier. In this way, the shallow layers could be better optimized and the network with few parameters could be directly obtained by the truncation of front-end network, avoiding the training of multiple networks. Besides [16], [50], SAD [51] extracts the attention maps from deeper layers, which act as the distillation target to guide the shallow layers. In addition, PS-KD [52] uses the linear combination of the hard-label and the past predictions at a certain epoch as the soft-label, which is adaptively adjusted as training progresses.

Inspired by the success of these works, self-distillation is also introduced into remote sensing scene classification. Duan et al. [45] develop the self-context distillation module (SCDM), in which the multiscale features from deeper layers are used to supervise the shallow parts of the model. CASD [43] constructs the input bag with images from different classes and extracts the class-aware information for distillation, which simultaneously alleviates the issue of intraclass diversity and interclass similarity in remote sensing images. FASDNet [46] effectively combines feature distillation and logits distillation, in which the feature augmentation pyramid module (FAPM) is designed to promote the feature representation and facilitate the knowledge transfer. EMSCNet [53] uses a dynamic dictionary to mine positive and negative pairs, based on which the contrastive learning strategy is used to calculate the similarity of each pair. Then, the similarity information is distilled into the backbone network to facilitate the feature learning.

The existing self-distillation methods enhance the performance of remote sensing scene classification. However, they

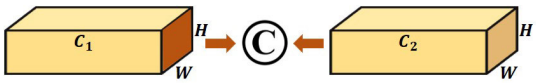


Fig. 4. Pipeline of the concatenation operation in our DiDi module.

ignore the disruptive factors and thorny noise present in the shallow layer features. Although VSDNet [7] proposes to establish a self-distillation framework based on the probabilistic model to learn category distributions in the label space, it is more elastic to fit the randomness in the inputs. Therefore, VSDNet could handle the uncertainty and idiosyncratic perturbation existing in the remote sensing images. However, VSDNet lacks explicit mechanisms to suppress noise factors, preventing it from establishing a robust learning paradigm for remote sensing images with perturbations. To solve this limitation and take a step further, we integrate a conditional DM into the self-distillation framework, explicitly addressing feature noise while enabling more effective knowledge transfer. To the best of our knowledge, this is the first implementation to incorporate a conditional DM into the self-distillation framework.

C. Diffusion Model

DM, also known as denoising diffusion probabilistic model [26], [54], uses a fixed Markov chain inspired by nonequilibrium thermodynamics to optimize the variation boundary of the likelihood function [55]. Due to the powerful capability of learning the prior distribution of images, the DM has gained widespread attention in recent years [56], [57].

Despite their remarkable performance, the pioneer DMs [26], [54], [58] exhibit low controllability, making it difficult to produce desired content. To address this challenge, additional information, such as category labels [24], text prompts [59], and geometric structures [27], [60], is introduced as conditions to guide the generation of the DM. For instance, ADM [24] is the first method to introduce conditional

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information into the DM, transforming denoising diffusion into a class-conditional task using a classifier-guided approach to balance the diversity and fidelity of the output. In addition, SR3 [61] incorporates high-level semantic features from low-resolution inputs to improve the fidelity and details of upsampled images, achieving high magnification image super-resolution. Beyond generation tasks, conditional DMs have also been applied to the perception tasks, such as object detection [27] and segmentation [60]. In specific, DiffusionDet [27] is the first framework to apply the DM to object detection, refining random generated boxes into precise predicted boxes through a denoising diffusion process. DiffusionDet demonstrates the effectiveness of using random boxes as valid object candidates in generative detection methods. Consequently, the conditional DM has increasingly become mainstream in the field of controllable image generation and also shown remarkable performance in perception tasks, exhibiting its significant potential and effectiveness in computer vision.

With the remarkable success in various vision tasks, DMs are increasingly applied to remote sensing scenarios [62]. For example, DMDC [63] designs a generative conditional DM tailored for remote sensing image super-resolution, effectively addressing the challenges of insufficient attention to small targets while enhancing the detail resolution. HyperLDM [25] constructs a hyperspectral remote sensing image generation framework based on latent DM by adopting false-color images as conditional information. In addition, LDS2AE [64] designs a diffusion autoencoder with several modality-specific decoders to learn multimodal homogeneous information, allowing the network to classify with arbitrary missing modalities through a single model. RSHazeDiff [65] introduces a unified Fourier-aware DM that effectively preserves image details and color fidelity in remote sensing image dehazing. In summary, these studies effectively verify that the DMs have strong modeling capabilities in different remote sensing tasks.

In this article, we further explore the application of DM in remote sensing scenario. To be more specific, we leverage the denoising capacity of the DM to remove the irrelevant disruptive factors within the shallow layer features, thereby enhancing the self-distillation for remote sensing scene classification.

III. Method

In this part, we elaborate on the details of our DSDNet. We first present the preliminary introduction about self-distillation and DM. Then, we exhibit the details of DiDi module. Finally, we introduce how to implement self-distillation with DiDi module and detail the whole pipeline of our DSDNet.

A. Preliminaries

1) Self-Distillation: For the image classification task, given an input image x, the one-hot ground-truth vector is denoted as $y = [0, 0, 1, \cdots, 0]^T \in R^K$, where K represents the total number of categories. The classification network $E(\cdot) \sim \theta$, composed of several convolutional blocks $\{\text{Conv_1}, \text{Conv_2}, \ldots, \text{Conv_}M\}$, generates multiple intermediate feature maps $S = \{S_1, S_2, \ldots, S_M\}$ through these convolutional blocks. Then, each feature map S_j is delivered to the corresponding classifier f_j to generate the logits e_j, based on which the prediction vector v_j is obtained

$$v_j^k = \frac{\exp\left(e_j^k\right)}{\sum_{k=1}^{K} \exp\left(e_j^k\right)} \tag{1}$$

in which k denotes the kth class among all the categories, e represents the logits produced by the fully connected layer, and q^k is the probability value of the kth class in the prediction vector q. As illustrated in Fig. 3, for $\{S_1, S_2, \ldots, S_M\}$, we obtain prediction vector $\{v_1, v_2, \ldots, v_M\}$ through $\{f_1, f_2, \ldots, f_M\}$. Afterward, the deep portion of the network is used as the teacher branch to guide the classifier in the shallow layers. Accordingly, the self-distillation loss is defined as [17]

$$\begin{aligned}\mathcal{L}_{kd} &= \sum_{i=1}^{N} v_{j,i}^{\text{tea}} \log\left(\hat{v}_{j,i}^{\text{tea}} / v_{k,i}^{\text{stu}}\right) \\ &= \sum_{i=1}^{N} v_{j,i}^{\text{tea}} \log\left(v_{j,i}^{\text{tea}}\right) - \sum_{i=1}^{N} v_{j,i}^{\text{tea}} \log\left(v_{k,i}^{\text{stu}}\right)\end{aligned} \tag{2}$$

where N is the batch size. j and k indicate the jth and kth prediction logits, respectively, in which $j = k + 1$ and $j, k \in \{1, 2, 3, \ldots, M\}$. In other words, v_j is generated at the deeper layer than v_k. The pair $< v_j^{\text{tea}}, v_k^{\text{stu}} >$ constitutes one <teacher-student> pair within the self-distillation framework. Through the cross-layer supervision in 2, the shallow part of the network is enhanced, which also adheres to the progressive learning strategy of neural networks.

2) Diffusion Model: DM [26], [54] is a class of probabilistic generative models, which define a Markov chain of forward-diffusion process by progressively adding noise to sample data, and then learn to reverse this process by predicting and removing the noise. Specifically, in the forward-diffusion process, the Gaussian noise is gradually added to the data with the control of a predefined noise variance schedule $\{\beta_t\}_{t=1}^T$ as

$$q(z_t|z_{t-1}) = \mathcal{N}\left(z_t; \sqrt{1-\beta_t} z_{t-1}, \beta_t \mathbf{I}\right) \tag{3}$$

where z_t and z_{t-1} are the noisy data at timestep t and $t-1 \in \{0, 1, \ldots, T\}$, respectively, and $\mathbf{I}$ represents the identity

Algorithm 1 Diffusion-training process

```
def train(tea_log, stu_log):
  """tea_log: logits from teacher branch"""
  """stu_log: logits from student branch"""
  """logits:[b, c, 1, 1]"""
  # corrupt tea_log
  t, eps = randint(0, T), normal(mean=0, std=1)
  tea_crpt = sqrt(alpha_cumprod(t)) * tea_log +
           sqrt(1 - alpha_cumprod(t)) * eps
  # predict noise
  noise_pred = diff_block(tea_crpt, stu_log, t)
  # calculate loss
  loss = L2_Distance(noise_pred, eps)
  return loss
```

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matrix. Meanwhile, $\{\beta_t \in (0,1)\}_{t=1}^{T}$ controls the degree of randomness, which satisfies $\beta_1 < \beta_2 < \cdots < \beta_T$. Then, we denote $\alpha_t = 1-\beta_t$, and thus $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s = \prod_{s=1}^{t}(1-\beta_s)$. Subsequently, through the iterations of t steps, z_t is expressed as

$$q(z_t|z_0) = \mathcal{N}\left(z_t; \sqrt{\bar{\alpha}_t} z_0, \left(1-\bar{\alpha}_t\right)\mathbf{I}\right) \tag{4}$$

where $z_0 \sim q(z_0)$ is the clean data without adding noise and $q(z_0)$ denotes the distribution of z_0. Typically, the reparameterization strategy is used to generate z_t as

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \epsilon_t \sqrt{1-\bar{\alpha}_t} \tag{5}$$

in which $\epsilon_t \in \mathcal{N}(\mathbf{0}, \mathbf{I})$ is sampled from a normal distribution at each step t and its intensity is defined by $(1-\bar{\alpha}_t)^{1/2}$.

During the training stage, Φ_θ is trained to predict the noise in z_t with respect to z_0. Then, $\mathcal{L}_{\text{diff}}$ is calculated to minimize the difference between the predicted noise and the ground-truth noise variable ϵ_t

$$\mathcal{L}_{\text{diff}} = ||\epsilon_t - \Phi_\theta(z_t, t)||^2. \tag{6}$$

Notably, the step t in the training process is randomly selected for each training iteration.

During the diffusion-inference process, the reverse denoising process starts from the initial noise $z_T \in \mathcal{N}(\mathbf{0}, \mathbf{I})$ and progressively removes the noise to recover z_0 through step-by-step denoising, which is defined as

$$p_\theta(z_{t-1}|z_t) = \mathcal{N}\left(z_{t-1}; \mu_\theta(z_t, t), \sigma_t^2 \mathbf{I}\right) \tag{7}$$

where σ_t^2 is controlled by β_t. Specifically, based on the derivation in [58], z_{t-1} is calculated as

$$z_{t-1} = \sqrt{\bar{\alpha}_{t-1}} z_0 + \sqrt{1-\bar{\alpha}_{t-1}-\sigma_t^2}\frac{z_t - \sqrt{\bar{\alpha}_t} z_0}{\sqrt{1-\bar{\alpha}_t}} + \sigma_t \epsilon_t \tag{8}$$

in which z_0 is calculated as

$$z_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(z_t - \sqrt{1-\bar{\alpha}_t}\Phi_\theta(z_t, t)\right). \tag{9}$$

By iteratively performing (8), the denoised z_0 could be generated. According to [58], to accelerate the diffusion-inference process, some denoising steps could be skipped as $z_T \rightarrow z_{T-\triangle} \rightarrow \cdots \rightarrow z_0$, where $\triangle$ is the interval.

B. Diffusion Distillation Module

In this part, we elaborate the details of our DiDi module. First, we introduce the conditional DM from two perspectives, diffusion-training and diffusion-inference. Then, we describe the knowledge transfer process within DiDi module.

Generally speaking, DiDi module is deployed between each selected <teacher-student> pair to facilitate knowledge distillation, with the shallow branch serving as the student model and the deeper branch as the teacher model. In the following part, for the sake of brevity, we denote a <teacher-student> prediction pair as $< \boldsymbol{q}, \boldsymbol{p} >$. Notably, it is significant to emphasize that both the diffusion-training and diffusion-inference processes are conducted concurrently during the training of DSDNet. Namely, the diffusion-training and diffusion-inference are jointly performed to promote the self-distillation and enhance the shallow layers of the network. Once the training is complete, all the DiDi modules will be discarded, leaving only the plain network to generate the final predictions.

Algorithm 2 Diffusion-inference process

```
def inference(stu_log, steps, ti=1):
 """stu_log: logits from student branch"""
 """steps: sample steps, ti: time intervals"""
 log_pred = normal(0, 1) # [b, c, 1, 1]
 for step in range(steps):
  # time intervals
  t_now = step
  t_next = step - ti
  # predicted noise at current step
  noise_pred = diff_block(log_pred, stu_log, t_now)
  # estimate logits at t_next
  log_pred = ddim_step(noise_pred, t_now, t_next)
 return log_pred
```

1) Diffusion-Training: For a specific DiDi module D_i, we use the prediction logits $\boldsymbol{q}$ from its corresponding teacher branch to train D_i during the diffusion-training. In other words, $\boldsymbol{q}$ serves as the learning target and is treated as the clean data $\boldsymbol{q}_0$ in the diffusion-training process. Algorithm 1 provides the pseudocode of diffusion-training process in our DiDi module. In each iteration, we randomly select the forward steps t from 0 to T and add Gaussian noise ϵ_t to $\boldsymbol{q}_0$. The noise intensity is controlled by α_t, which adopts the monotonically decreasing schedule for α_t in different time steps t. By doing so, we will obtain the noised feature $\boldsymbol{q}_t$ corresponding to different steps in the diffusion-training process.

Afterward, we use the diffusion block $\Phi_\theta(\boldsymbol{q}_t, t|\boldsymbol{p})$ to predict the noise conditioned on the student logits $\boldsymbol{p}$. The diffusion block $\Phi_\theta(\cdot)$ is composed of three sets of "Convolution-BN-ReLU" operations, followed by an additional convolutional layer to generate the output. As depicted in Fig. 3, we concatenate $\boldsymbol{q}_t$ with $\boldsymbol{p}$ as the condition, and then predict the noise intensity at the current step t. The concatenation operation is depicted in Fig. 4, which combines two features along the channelwise dimension. Subsequently, according to (6), we calculate the l_2 distance between the predicted noise $\Phi_\theta(\boldsymbol{q}_t, t|\boldsymbol{p})$ and ground-truth noise ϵ_t to calculate $\mathcal{L}_{\text{diff}}$. Furthermore, we explain the training objective in terms of a variational lower bound. During the diffusion-training stage, we consider the forward diffusion process as a fixed approximate posterior to the diffusion-inference process. This approach allows us to derive the following variational lower bound on the marginal log-likelihood:

$$\begin{aligned}&\mathbb{E}_{(\boldsymbol{p},\boldsymbol{q}_0)} \log p_\theta\left(\boldsymbol{q}_0 \mid \boldsymbol{p}\right) \\ &\geq \mathbb{E}_{\boldsymbol{p},\boldsymbol{q}_0}\mathbb{E}_{q\left(\boldsymbol{q}_{1:T}|\boldsymbol{q}_0\right)}\left[\log p\left(\boldsymbol{q}_T\right) + \sum_{t\geq 1}\log\frac{p_\theta\left(\boldsymbol{q}_{t-1} \mid \boldsymbol{q}_t, \boldsymbol{p}\right)}{q\left(\boldsymbol{q}_t \mid \boldsymbol{q}_{t-1}\right)}\right].\end{aligned} \tag{10}$$

2) Diffusion-Inference: In the diffusion-inference phase, we aim to eliminate the noise from student logits $\boldsymbol{p}$ and

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Algorithm 3 Training of DSDNet

Initialization: $E(x)$, Diffusion-training, Diffusion-inference
Input: x∼image, y∼label

repeat
  $f_1, f_2, f_3, f_4 = E(x)$;
  $pred = \text{Classifier}(f4)$, $loss_{\text{ce}} = \text{CE-loss}(pred, y)$;
  $loss_{\text{diff}} = \text{Diffusion-training}(f_j, f_{(j+1)})$;
  $log\text{-}pred_j = \text{Diffusion-inference}(f_j, f_{(j+1)}, ti)$;
  $loss_{kd} = \text{KD-loss}(\text{log-pred}_j, \text{log-pred}_{j+1})$;
  $LOSS_{all} = loss_{\text{ce}} + loss_{\text{diff}} + loss_{kd}$;
  Minimize $LOSS_{all}$
until *converged;*

Algorithm 4 Inference of DSDNet

Given: $E(x)$, Diffusion-training, Diffusion-inference
Input: $x \sim$ image
  $f_1, f_2, f_3, f_4 \leftarrow E(x)$
  $pred \leftarrow \text{Classifier}(f_4)$
return *pred*

generate the refined representation via the DM $\Phi_\theta(\cdot)$ with a predefined sampling step T'. The inference procedure is outlined in Algorithm 2. In the first step of the sampling, the randomly initialized logits $\hat{\boldsymbol{p}}_{T'}$, concatenated with the student logits $\boldsymbol{p}$, are delivered into the diffusion block $\Phi_\theta(\cdot)$ to predict the noise intensity $\epsilon_{T'}$ at the current step. Then, based on ϵ'_T, we compute the estimated denoised logits $\hat{\boldsymbol{p}}_{T'-1}$. In subsequent steps, the estimated logits from the previous step, along with $\boldsymbol{p}$, are sent into the diffusion block for further denoising. Therefore, we define the entire diffusion-inference process in terms of isotropic Gaussian conditional distributions $p_\theta(\boldsymbol{q}_{t-1} \mid \boldsymbol{q}_t, \boldsymbol{p})$, which can be described as

$$p_\theta(\boldsymbol{q}_{0:T'} \mid \boldsymbol{p}) = p(\boldsymbol{q}_{T'}) \prod_{t=1}^{T'} p_\theta(\boldsymbol{q}_{t-1} \mid \boldsymbol{q}_t, \boldsymbol{p}) \tag{11}$$

$$p(\boldsymbol{q}_{T'}) = \mathcal{N}(\boldsymbol{q}_{T'}; \boldsymbol{0}, \boldsymbol{I}) \tag{12}$$

$$p_\theta(\boldsymbol{q}_{t-1} \mid \boldsymbol{q}_t, \boldsymbol{p}) = \mathcal{N}(\boldsymbol{q}_{t-1}; \mu_\theta(\boldsymbol{p}, \boldsymbol{q}_t, t), \sigma_t^2 \boldsymbol{I}). \tag{13}$$

Ultimately, the refined representation $\hat{\boldsymbol{p}}_0$ is generated through T' steps. Notably, following the success of DDIM [58], the sampling steps T' in DSDNet are fewer than the diffusion forward steps T, which accelerates the sampling process and improves the efficacy.

3) Knowledge Transfer: Through diffusion-inference, the generated $\hat{\boldsymbol{p}}_0$ is treated as the refined representation of student logits, which has been filtered to remove the disruptive factors. Subsequently, the distillation loss $\mathcal{L}_{kd}$ is applied between <teacher, student> pair $< \boldsymbol{q}, \hat{\boldsymbol{p}}_0 >$ as defined in 2. Since the distillation is only operated in the training of DSDNet, it is crucial to perform the diffusion-inference process concurrently to refine the features of the student branch, thereby enhancing the effectiveness of distillation. Therefore, as outlined in Algorithm 3, diffusion-training and diffusion-inference are jointly conducted throughout the training phase.

C. Framework of DSDNet

As illustrated in Fig. 3, we divide the backbone network into four stages, i.e., $M = 4$. Consequently, we obtain four intermediate feature maps $\{S_1, S_2, S_3, S_4\}$. Among them, S_4 is sent into the final classifier f_4 to generate the final prediction v_4, which is supervised by the cross-entropy loss

$$\mathcal{L}_{\text{ce}} = \frac{1}{N} \sum_{i=1}^{N} y_i \log(v_i) \tag{14}$$

in which N represents the batch size and y_i denotes the ground truth of the ith sample.

Meanwhile, as depicted in Fig. 3, we formulate four <teacher, student> pairs, each of which is delivered into the DiDi module to perform self-distillation. By doing so, the knowledge from deeper layers is squeezed into the shallow parts, enhancing the gradient of the shallow layers and thereby promoting their representation capacity. In addition, through the DMs within the DiDi module, the disruptive noise in the student branch is removed, further facilitating the effectiveness of the distillation process.

Finally, combining (2), (6), and (14), the total loss of DSDNet is calculated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}} + \mathcal{L}_{kd} + \mathcal{L}_{\text{ce}}. \tag{15}$$

The inference pipeline is presented in Algorithm 4. During the inference of DSDNet, all the auxiliary classifiers and DiDi modules will be discarded, leaving only the backbone network used for prediction. In other words, the DMs are used exclusively during training. Once training is complete, the backbone network can extract discriminative features independently, without adding additional parameters.

IV. Experiments

To evaluate the performance of the proposed method, we conducted extensive experiments on four challenging remote sensing scene classification datasets, NWPU-RESISC45 [66], AID [67], OPTIMAL-31 [68], and UC-Merced [69]. The experimental results demonstrate that DSDNet achieves state-of-the-art accuracy, showcasing its remarkable effectiveness and robustness in remote sensing scene classification.

A. Dataset

In this part, we briefly introduce the four datasets used in our experiments. In Fig 5, we depict some selected samples along with their corresponding category names.

1) NWPU-RESISC45 [66]: The NWPU-RESISC45 is one of the largest datasets for remote sensing scene classification, comprising 31 500 images across 45 different scene categories. The spatial resolution of most images ranges from 30 to 0.2 m/pixel. Following strategies in prior works [70], [71], [72], we conduct experiments with two different training ratios. Specifically, 10%–20% of the dataset are used as the training set, respectively. The corresponding remaining 90%–80% are used as the test set.

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Fig. 5. We present sample images from the four remote sensing scene classification datasets used in our experiments. From top to bottom, the datasets are NWPU-RESISC45, AID, OPTIMAL-31, and UC-Merced, respectively. The corresponding category labels are provided below each image.

2) AID [67]: The aerial image dataset (AID) is a comprehensive remote sensing scene classification dataset containing 10 000 samples across 30 different scene categories. The resolution of each image varies from 8 to 0.5 m. In our experiments, we adopt the commonly used training ratios of 20%–50%, with the remaining 80%–50% of the data used for testing [32].

3) OPTIMAL-31 [68]: The OPTIMAL-31 includes 31 categories, each of which contains 60 images. Following the procedure outlined in [68], we use 80% of the dataset for training and the remaining 20% for evaluation. Notably, the size of OPTIMAL-31 is much smaller compared with the other three datasets involved in the experiments, which significantly increases the challenges for accurate classification.

4) UC-Merced [69]: The UC-Merced dataset is composed of 21 scene classes, each with 100 images. The spatial resolution of UC-Merced is 0.3 m/pixel. Consistent with previous works [73], [74], [75], we perform two groups of experiments with the training ratio of 50% and 80%, respectively. Then, the remaining samples are used for evaluation.

B. Experimental Settings

In the experiments, we adopt the root mean square propagation (RMSProp) [83] as the optimizer, in which the weight decay is set to 1e-5. The initial learning rate is set to 1e-4 and it is reduced to a minimum of 1e-6 through the cosine decay strategy for gradual adjustment. In the training stage, we enhance the input diversity through augmentations such as random horizontal flipping, random rotation, and color jitter. Images are resized to 448 $\times$ 448. In the test, we resize the short edge of the image to 448, while preserving the aspect ratio, followed by a center-crop to obtain the 448 $\times$ 448 patch for evaluation. In addition, in the diffusion-training process, the maximum forward step T is set to 1000. During the diffusion-inference process, following the success of [58], some denoising steps are skipped and the sampling step T' is set to 5 to accelerate the speed. Notably, to mitigate the effect of randomness and ensure robust results, we repeat the main experiments five times and report the average result.

As for the evaluation metric, we use overall accuracy to measure the performance of the proposed DSDNet, which is defined as the ratio of correctly predicted samples to the total number of test samples. In addition, we present the confusion matrix to further analyze the classification errors across different categories. It is worth noting that in the confusion matrix, we only report the accuracy values greater than 0.001. Therefore, the sum of some rows in the illustrated confusion matrix may not equal 1.0.

C. Network Details

To assess the compatibility of the proposed method, we evaluate the performance of DSDNet with various backbone architectures including VGG-16 [11], ResNet-34 [12], ResNet-50 [12], and ViT-Base-16 [13], respectively. The experimental results show that our method consistently improves performance across these different backbones.

In our DSDNet, we incorporate auxiliary classifiers at different intermediate layers of each backbone in addition to the final classifier located at the end of the network. For VGG-16, auxiliary classifiers are attached at the relu2_3, relu3_3, and relu4_3 layers. In the case of ResNet-34 and ResNet-50, auxiliary classifiers are deployed after conv2_x, conv3_x, and conv4_x, respectively. As for ViT-Base-16, we attach auxiliary classifiers at encoder_layer_2, encoder_layer_5, and encoder_layer_8, respectively. Based on these auxiliary classifiers, along with the final classifier, we construct <teacher,

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TABLE I
ACCURACY COMPARISON TO DIFFERENT METHODS ON THE NWPU-RESISC45 DATASET

Method	Training ratio	
	10%	20%
VGGNet-16 based Methods		
VGG-16 [11]	88.17±0.32	91.62±0.33
HW-CNN-VGG16 [71]	-	94.38±0.16
KFBNet-VGG16 [76]	90.27±0.02	92.54±0.03
CapsNet-VGG16 [77]	85.08±0.13	89.18±0.14
MG-CAP-VGG16 [78]	90.83±0.12	92.95±0.13
ACNet-VGG16 [79]	91.09±0.13	92.42±0.16
ResNet based Methods		
ResNet-34 [12]	90.24±0.21	92.78±0.18
ResNet-50 [12]	90.76±0.11	93.11±0.08
EMTCAL-ResNet34 [41]	91.63±0.19	93.65±0.12
CDLNet-ResNet50 [37]	92.28±0.25	94.50±0.19
ViT-B based Methods		
ViT-B [13]	91.59±0.19	93.90±0.20
EMSCNet-ViT-B [53]	93.58±0.22	95.37±0.07
CDLNet-ViT-B [37]	93.43±0.18	95.30±0.03
Methods based on other architecture		
CapsNet-Inception-v3 [77]	89.03±0.21	92.60±0.11
LGRIN [75]	91.91±0.15	94.43±0.16
Hydra [72]	92.44±0.34	94.51±0.21
Triplet Network [80]	-	92.33±0.20
SCViT [81]	92.72±0.04	94.66±0.10
HHTL [40]	92.07±0.44	94.21±0.09
Self-Distillation based Methods		
FASDNet-ResNet34 [46]	92.89±0.13	94.95±0.12
CASD-ResNet50 [43]	92.28±0.23	94.75±0.14
EMSCNet-ResNet50 [53]	92.16±0.07	94.08±0.20
CASD-ViT-B [43]	93.12±0.12	95.52±0.16
Ours		
DSDNet-VGG16	90.57±0.04	93.47±0.04
DSDNet-ResNet34	92.97±0.03	94.81±0.03
DSDNet-ResNet50	93.23±0.03	95.03±0.05
DSDNet-ViT-B	93.74±0.05	95.61±0.07

TABLE II
ACCURACY COMPARISON TO DIFFERENT METHODS ON AID

Method	Training ratio	
	20%	50%
VGGNet-16 Based Methods		
VGG-16 [11]	92.63±0.23	94.59±0.19
HW-CNN-VGG16 [71]	-	96.98±0.33
GBNet-VGG16 [74]	92.20±0.23	95.48±0.12
MG-CAP-VGG16 [78]	93.34±0.18	96.12±0.12
CapsNet-VGG16 [77]	91.63±0.19	94.74±0.17
FACNN-VGG16 [82]	-	95.45±0.11
KFBNet-VGG16 [76]	94.27±0.02	97.19±0.07
ACNet-VGG16 [79]	93.33±0.29	95.38±0.29
ResNet Based Methods		
ResNet-34 [12]	93.68±0.21	95.27±0.23
ResNet-50 [12]	94.16±0.22	95.64±0.18
EFPNet-ResNet34 [73]	94.02±0.21	94.50±0.30
EMTCAL-ResNet34 [41]	94.69±0.14	96.41±0.23
CDLNet-ResNet50 [37]	95.47±0.11	97.19±0.11
ViT-B Based Methods		
ViT-B [13]	94.90±0.29	96.49±0.18
CDLNet-ViT-B [37]	96.07±0.16	97.31±0.12
EMSCNet-ViT-B [53]	96.02±0.18	97.35±0.17
Methods based on other architecture		
LGRIN [75]	94.74±0.23	97.65±0.25
CapsNet-Inception-v3 [77]	93.79±0.13	96.32±0.12
SCViT [81]	95.56±0.17	96.98±0.16
HHTL [40]	95.62±0.13	96.88±0.21
Self-Distillation based Methods		
FASDNet-ResNet34 [46]	96.05±0.13	97.84±0.12
CASD-ResNet50 [43]	95.72±0.13	96.96±0.16
EMSCNet-ResNet50 [53]	95.13±0.10	96.96±0.10
CASD-ViT-B [43]	96.18±0.20	97.64±0.11
Ours		
DSDNet-VGG16	95.37±0.05	97.32±0.05
DSDNet-ResNet34	96.04±0.01	97.41±0.02
DSDNet-ResNet50	96.23±0.04	97.68±0.05
DSDNet-ViT-B	96.81±0.07	98.19±0.09

student> pair according to the pipeline in Fig. 3. The DiDi module is then applied between each <teacher, student> pair to distill the knowledge from the deeper layers into shallow parts of the network progressively.

D. Performance Comparison

To assess the effectiveness of our approach, we conduct comprehensive experiments and benchmark DSDNet against various remote sensing scene classification networks. The results are presented in Tables I–IV, respectively. As depicted, each table is organized into six parts according to the rows. From the top to bottom, the first three parts detail the accuracy of models based on VGG-16, ResNet-34(50), and ViT-B, respectively. The fourth part summarizes the accuracy of methods using some other backbones or established on its unique structure, including both the CNN-based and transformer-based networks. The fifth part highlights the performance of some representative self-distillation-based methods. The last row reflects the accuracy of our DSDNet. Generally speaking, DSDNet consistently enhances performance across various backbones and establishes new state-of-the-art results. The comprehensive analysis for each dataset is provided below.

TABLE III
ACCURACY COMPARISON TO DIFFERENT METHODS ON OPTIMAL-31

Method	Training Ratio: 80%
VGG-16 Based Methods	
VGG-16 [11]	92.50±0.41
ARCNet-VGG16 [68]	92.70±0.35
GBNet-VGG16 [74]	93.28±0.27
KFBNet-VGG16 [76]	95.12±0.13
ResNet Based Methods	
ResNet-34 [12]	93.91±0.37
ResNet-50 [12]	93.96±0.14
Methods based on other architecture	
ARCNet-AlexNet [68]	85.75±0.35
ViT-B Based Methods	
ViT-B [13]	94.72±0.22
Self-Distillation based Methods	
CASD-ResNet50 [43]	95.58±0.23
CASD-ViT-B [43]	96.05±0.26
Ours	
DSDNet-VGG16	95.19±0.17
DSDNet-ResNet34	96.83±0.13
DSDNet-ResNet50	96.56±0.21
DSDNet-ViT-B	97.02±0.20

1) Results on NWPU-RESISC45: As shown in Table I, DSDNet delivers the best performance with an accuracy of

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TABLE IV
ACCURACY COMPARISON TO DIFFERENT METHODS ON UC MERCED

Method	Training ratio	
	50%	80%
VGGNet-16 Based Methods		
VGG-16 [11]	96.18±0.38	98.11±0.32
ARCNet-VGG16 [68]	96.88±0.21	98.23±0.13
GBNet-VGG16 [74]	97.05±0.19	98.57±0.48
FACNN-VGG16 [82]	-	98.81±0.24
CapsNet-VGG16 [77]	95.33±0.18	98.81±0.22
MG-CAP-VGG16 [78]	-	99.00±0.10
KFBNet-VGG16 [76]	-	99.76±0.24
ResNet Based Methods		
ResNet-34 [12]	96.84±0.25	98.28±0.26
ResNet-50 [12]	96.88±0.21	98.23±0.13
EFPNet-ResNet34 [73]	96.19±0.13	99.14±0.22
EMTCAL-ResNet34 [41]	98.67±0.16	99.57±0.28
CDLNet-ResNet50 [37]	98.72±0.17	99.66±0.11
Methods based on other architecture		
CapsNet-Inception-v3 [77]	97.59±0.16	99.05±0.24
LGRIN [75]	98.61±0.22	98.97±0.31
ViT-B Based Methods		
ViT-B [13]	97.48±0.39	99.01±0.21
CASD-ViT-B [43]	99.07±0.09	99.70±0.11
SCViT [81]	98.90±0.19	99.57±0.31
HHTL [40]	98.87±0.28	99.48±0.28
EMSCNet-ViT-B [53]	99.10±0.20	99.71±0.17
CDLNet-ViT-B [37]	99.06±0.15	99.71±0.09
Self-Distillation based Methods		
FASDNet-ResNet34 [46]	98.49±0.18	99.67±0.18
CASD-ResNet50 [43]	98.58±0.13	99.62±0.16
EMSCNet-ResNet50 [53]	98.70±0.46	99.44±0.16
DSDNet-VGG16	98.12±0.11	99.58±0.13
DSDNet-ResNet34	98.98±0.08	99.67±0.08
DSDNet-ResNet50	99.03±0.07	99.71±0.06
DSDNet-ViT-B	99.11±0.13	99.75±0.12

93.74 ± 0.05 and 95.61 ± 0.07 for 10%–20% training ratio, respectively. Specifically, through the self-distillation enabled by DiDi module, DSDNet gains 2.4%, 2.73%, 2.47%, and 2.15% performance enhancements on VGG-16, ResNet-34, ResNet-50, and ViT-Base on 10% training ratio. In addition, in Fig. 6, we display the confusion matrix of DSDNet with 10% training ratio.

2) Results on AID: The experimental results on AID are presented in Table II, in which DSDNet equipped with ViT-B achieves the highest accuracy on both 20%–50% training ratio. Meanwhile, similar to the results on NWPU-RESISC45, DSDNet obtains consistent improvements when equipped with four different backbone networks, demonstrating the significant advantages of the proposed DiDi module. Furthermore, Fig. 7 displays the confusion matrix for DSDNet using the ResNet-50 backbone on the 20% training ratio, providing detailed insights into its classification performance.

3) Results on OPTIMAL-31: We present the experimental results on OPTIMAL-31 in Table III. As listed, 80% of the dataset is used to train the model and DSDNet delivers the best performance with the accuracy of 97.02 ± 0.20. Compared with the results on the other three datasets, DSDNet exhibits higher variance on OPTIMAL-31, likely due to the scarcity of the dataset, as it contains only 60 samples per category which is significantly fewer than the other datasets.

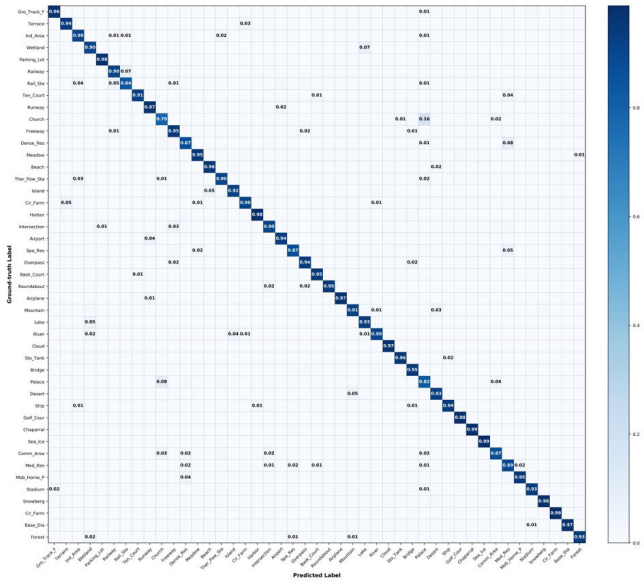


Fig. 6. Confusion matrix of DSDNet based on ResNet-50 over NWPU-RESISC45 under the 10% training ratio.

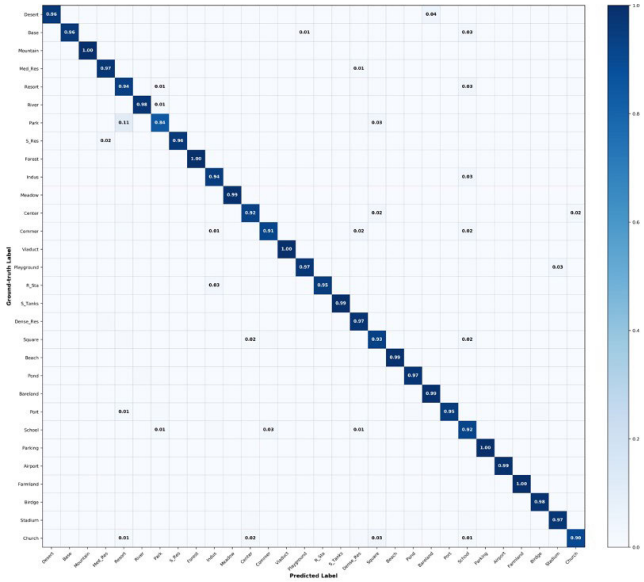


Fig. 7. Confusion matrix of DSDNet based on ResNet-50 over AID under the 20% training ratio.

Despite this, DSDNet still achieves consistent performance improvements, with notable accuracy increases on VGG-16, ResNet-34, ResNet-50, and ViT-Base, respectively. Fig. 8 displays the confusion matrix for DSDNet on OPTIMAL-31, illustrating its performance across different categories.

4) Results on UC-Merced: The experimental results of UC-Merced are listed in Table IV. It is evident that DSDNet delivers the remarkable performance. In particular, 1.94%, 2.14%, 2.15%, and 1.63% accuracy increases are observed on four backbone networks. The confusion matrix is illustrated in Fig. 9 to provide further details of the classification results.

E. Ablation Study

In this section, we conduct an extensive ablation study to verify the effectiveness of different components in our DSDNet. Unless otherwise specified, all the experiments in the

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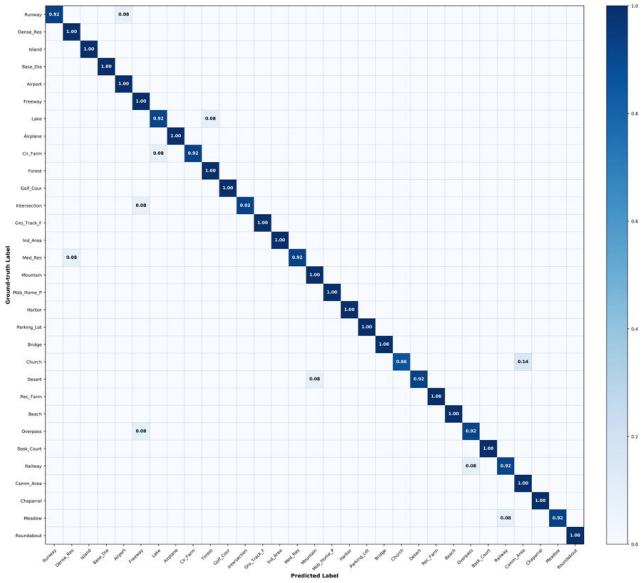


Fig. 8. Confusion matrix of DSDNet based on ResNet-50 over OPTIMAL-31 under the 80% training ratio.

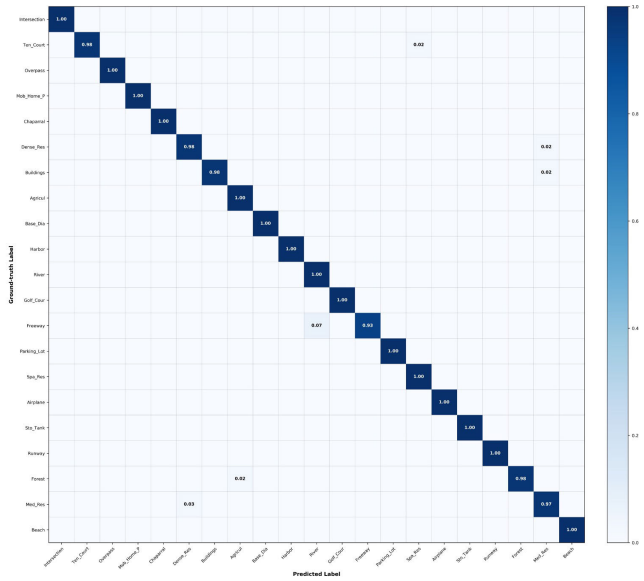


Fig. 9. Confusion matrix of DSDNet based on ResNet-50 over UC-Merced under the 50% training ratio.

TABLE V

ACCURACY COMPARISON WHEN DSDNET ADOPTS DIFFERENT DISTILLATION STRATEGIES

Distillation Strategy	Accuracy
Feature Distillation	92.99±0.05
Logits Distillation	93.23±0.03

ablation study are performed on the NWPU-RESISC45 dataset with a 10% training ratio, using ResNet-50 as the backbone network.

1) Effectiveness of Logits Distillation: There are two different strategies in the distillation field: i.e., logits distillation and feature distillation. The primary difference between the two strategies lies in the signals used for knowledge transfer. Our DSDNet uses logits distillation. To verify the effectiveness and robustness of this approach, we replace logits distillation with feature distillation, in which the feature maps S_1, S_2, S_3, and S_4 are delivered into DiDi module for diffusion learning. Overall, as listed in Table V, there is no significant difference when using the two different distillation strategies, reflecting the robustness of our method. Specifically, the best performance is achieved with logits distillation. We attribute this phenomenon to the "dark knowledge" encapsulated in the logits, which conveys information about interclass similarities, providing more useful knowledge than the intermediate features [17].

TABLE VI

ACCURACY COMPARISON WHEN DSDNET USES DIFFERENT NUMBERS OF <TEACHER-STUDENT> PAIRS TO PERFORM SELF-DISTILLATION. THE CONFIGURATION SHOWS THE POSITIONS WHERE WE ATTACH THE INTERMEDIATE CLASSIFIERS

Configuration	Accuracy
conv1_x, conv2_x, conv3_x, conv4_x	93.16±0.03
conv3_x, conv4_x	92.63±0.06
conv2_x, conv3_x	92.45±0.10
conv4_x	91.95±0.09
conv2_x, conv3_x, conv4_x	93.23±0.03

TABLE VII

ACCURACY COMPARISON WHEN DSDNET EQUIPPED WITH OR WITHOUT DM

Configuration	Accuracy
without Diffusion Model	92.37±0.12
with Diffusion Model	93.23±0.03

2) Effectiveness of Distillation Pairs: The number of <teacher-student> pairs is an important hyperparameter in self-distillation. Here, we carry out experiments to evaluate the influence of different numbers of <teacher-student> pairs. The experimental results are presented in Table VI. In this table, the configuration shows the position where we attach the intermediate classifier. The logits from adjacent classifiers are input into the DiDi module for self-distillation. According to the results, DSDNet achieves optimal performance when incorporating three DiDi modules among three adjacent <teacher-student> pairs. Notably, introducing an additional auxiliary classifier after conv1_x does not improve classification accuracy, suggesting that three auxiliary classifiers are sufficient to enhance the gradient in shallow layer features. Meanwhile, the information in the very shallow layers may be too raw and contain excessive irrelevant information. Therefore, incorporating conv1_x into the self-distillation does not further improve the performance. Moreover, removing one or more auxiliary classifiers along with their corresponding DiDi modules results in the significant performance drop. We hypothesize that in this scenario, the remaining auxiliary classifiers are somewhat sparse, failing to provide adequate gradient enhancement for the backbone network.

3) Effectiveness of DM: The DiDi module is a critical component in our DSDNet, where diffusion learning is implemented. In this part, to validate the effectiveness of the DM, we replace the DiDi module with a general knowledge transfer strategy, using only the distillation loss $\mathcal{L}_{kd}$ to transfer knowledge from deep layers to shallow layers. In this condition, the denoising effect of the DM is removed, making the network configuration similar to HSDNet [17]. The experimental results in Table VII indicate that the classification

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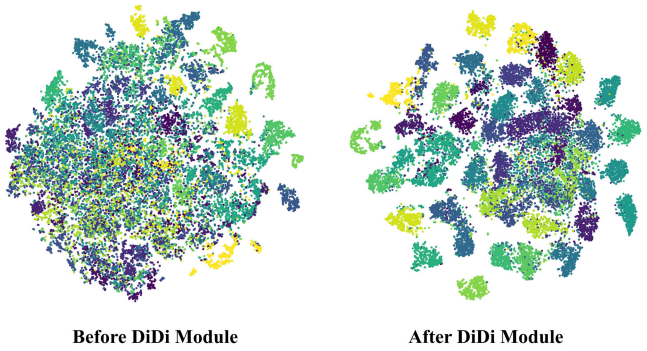


Fig. 10. Distribution of feature representations before and after passing through the DiDi module.

accuracy decreases by 0.86% when the DiDi module is replaced. We attribute this result to the denoising capability of the DM in refining low-quality feature representations. During the training of DSDNet, the diffusion-inference process filters out disruptive factors in shallow layer features used for self-distillation. In this way, the effectiveness of knowledge transfer in self-distillation is promoted and the shallow parts of the network are strengthened, which thereby improves the classification accuracy.

Moreover, to enhance the interpretability of the proposed method, we visualize the feature distribution before and after the DiDi module. Specifically, we use ResNet-50 as the backbone network and extract features from the auxiliary classifier attached at Conv4_x before and after passing through the DiDi module. Then we illustrate the feature distribution in Fig. 10. We can observe that the features after the DiDi module exhibit a more structured and organized distribution compared with those before, demonstrating the effectiveness of the denoising process.

F. Robustness Evaluation

In this part, we evaluate the robustness of the proposed method by introducing distortions to the images in the NWPU-RESISC45 dataset, specifically by adding noise and darkening the images. We then use the proposed DSDNet, which uses ResNet-50 as the backbone and is trained on the NWPU-RESISC45 under the training ratio of 10%, to perform zero-shot evaluation on these distorted images. The experimental details and results are presented below.

1) Image Distortion Approaches: Here, we detail the methods used to introduce distortions into the images. Generally speaking, we use two different techniques to degrade the images, i.e., noise addition and brightness reduction. Examples of selected original images alongside their corresponding distorted versions are illustrated in Fig. 11.

a) Noise addition: To evaluate the robustness of our model against random visual noise, we introduce Gaussian noise to the images, which is generated with a mean of 0 and a standard deviation of 15. The noisy image is produced by adding the generated noise to the original image. By doing so, we introduce a notable yet manageable level of noise, simulating conditions like digital or sensor noise.

b) Brightness reduction: To simulate low-light conditions, such as those encountered during dusk or in shadowed areas, we reduce the brightness of the remote sensing images by a factor of 0.5. These dark images help us evaluate the capacity of model in maintaining accuracy under the challenging lighting conditions that are typical in remote sensing scenarios.

2) Results of Robustness Evaluation: The results of robustness evaluation are listed in Table VIII. As shown, DSDNet exhibits remarkable robustness against different distortions, which outperforms plain ResNet-50 significantly. Specifically, DSDNet shows an 11.15% accuracy decrease in noisy images, which is smaller than the 16.10% decrease for plain ResNet-50. Meanwhile, when tested on low-light images, DSDNet exhibits a much smaller performance drop of only 5.35%, in contrast to the 17.74% decrease observed for ResNet-50.

In addition, we compare the robustness of DSDNet without the DM, while retaining the self-distillation structure. As shown in Table VIII, compared with the plain network, self-distillation could enhance the robustness with smaller performance drop. However, comparing the second and third rows in Table VIII, it is clear that DSDNet equipped with the DM exhibits superior robustness. During the training phase, the denoising capability of the DM effectively removes disruptive factors in shallow layer features, ensuring that self-distillation is not affected by these irrelevant factors and enhancing the robustness of network to such disruptions. Consequently, the feature extraction capacity of network when dealing with distorted images is improved.

G. Evaluation of Downstream Applicable

To evaluate the generalization capacity of the proposed DiDi module, we transfer DiDi module to the remote sensing segmentation task. Specifically, we conduct experiments based on the DeepLab v3+ with ResNet-50 as the backbone network. The experiments are performed on the iSAID dataset [84], which is one of the largest remote sensing segmentation datasets. During the experiments, we maintain the original structure unchanged and deploy auxiliary decoders (i.e., segmentation heads) at the different intermediate layer. Specifically, we attach segmentation heads after the second-to-last and third-to-last convolutional blocks in the DeepLab backbone. The segmentation map from each segmentation head is then used as the target to supervise the preceding one. Meanwhile, during the inference stage, all the auxiliary segmentation heads will be removed and only the final one will be used to produce the final prediction. In other words, the segmentation framework follows a similar design to the proposed DSDNet. In this way, we establish the self-distillation framework for the segmentation network, which enables the progressive enhancement of shallow layers and enhances the capacity of feature extraction.

The experimental results are listed in Table IX. As shown, when equipped with DiDi module, the DeepLab v3+ achieves higher mIOU performance with a 1.5% mIOU increase. The experimental results demonstrate that through knowledge transfer with denoising enhancement, the proposed DiDi module could facilitate the feature extraction in the segmentation

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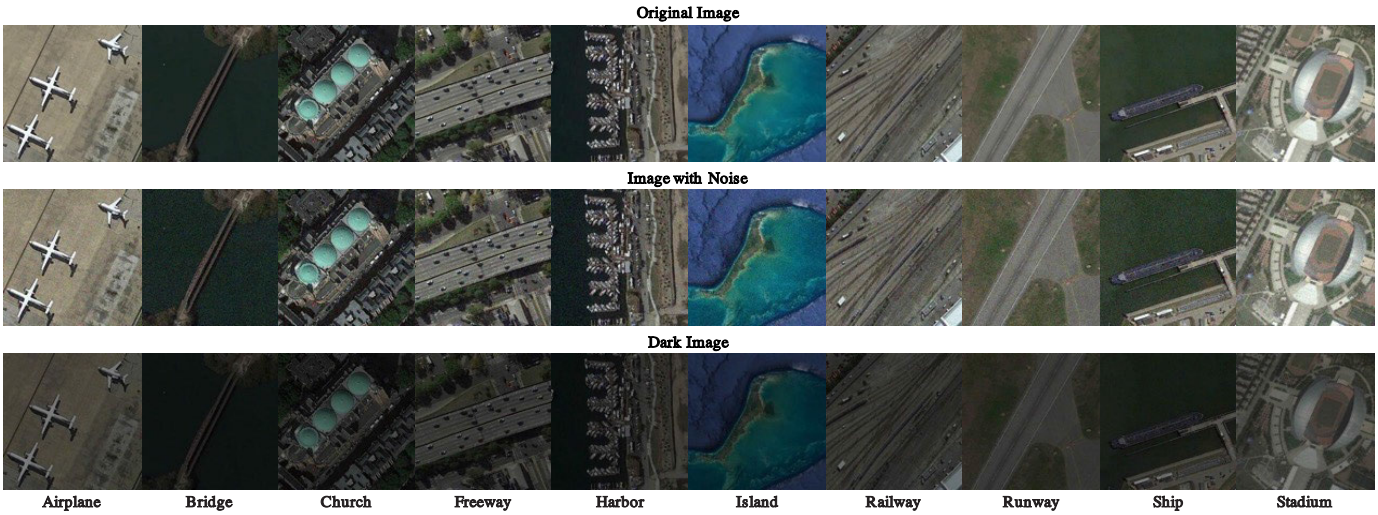


Fig. 11. We present a series of example images from the NWPU-RESISC45 dataset, along with their corresponding distorted versions. The original images are illustrated in the first row, followed by images with noise and dark images. The category of each image is provided in the last row. Best viewed in zoomed-in view.

TABLE VIII
ACCURACY COMPARISON OF DIFFERENT METHODS WHEN FACING VARIOUS DISTORTED IMAGES ON NWPU-RESISC45

Method	Original // Image	Image with Noise		Image with Low-Light	
		Acc ↑	Delta ↓	Acc ↑	Delta ↓
ResNet-50	90.76±0.11	74.66±0.52	16.10	73.02±0.68	17.74
DSDNet w/o Diffusion	92.37±0.12	78.93±0.34	13.44	78.75±0.49	13.62
DSDNet-Res50	93.23±0.03	82.08±0.32	11.15	87.88±0.46	5.35

TABLE IX
PERFORMANCE COMPARISON OF DEEPLAB V3+ WITH OR WITHOUT DIDI MODULE

Configuration	mIOU(%)
DeepLab v3+	59.3
DeepLab v3+ with DiDi module	60.8

network and demonstrates remarkable generalization capacity when applied to downstream tasks.

V. CONCLUSION

Remote sensing images present significant challenges due to small interclass variations and large intraclass variations. Obtaining accurate classification performance hinges on extracting powerful feature representations. To achieve this goal, in this article, we propose DSDNet, which enhances the effectiveness of self-distillation by leveraging the denoising capability of the DM. To the best of our knowledge, this is the first integration of a DM with self-distillation. Specifically, we treat shallow features as noisy versions of deep layer features. To mitigate the disruptive factors within shallow layer features, we use conditional DMs to remove the noise in the student features. As a result, the raw features from the shallow layers are refined, allowing critical classification information to be more effectively distilled into the shallow part of the network. In this way, the feature extraction capacity of the shallow layers is enhanced, thereby leading to a more discriminative feature representation and bringing the more accurate classification performance. DSDNet is particularly well-suited for remote sensing scene classification, where perturbations in remote sensing images often introduce substantial noise in shallow layer features. Extensive experiments on four challenging datasets demonstrate that DSDNet consistently improves performance across various backbone networks and achieves state-of-the-art results.

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